HUMBER POLYTECHNIC | Faculty of Applied Sciences & Technology

ENGI 4000

CAPSTONE PROJECT 1

CHAPTER 4

PRODUCT SPECIFICATIONS





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Table of Contents

Contents

Executive Summary 2
Section 1: Preparing the List of Metrics 3
1.1 Metrics of customer Needs 4
1.2 Needs-to-Metrics Mapping 5
Section 2: Develop Technical Models of the Product 6
2.1 Model 1: Carriage Acceleration and Motor Torque Sizing 7
2.2 Model 2: Repeatability Error Budget 8
2.3 Model 3: Vibration Settling Time 9
2.4 Model 4: Emergency-Stop Distance and Response Time 10
Section 3: Establishing Ideal and Marginally Acceptable Target Values 11
Reflection and Justification of Marginal and Ideal Metric Values for the CMR 12
References 15

Executive Summary

This chapter focuses on turning the eight customer needs identified into measurable specifications for the camera system. The chapter is divided into three sections.

Section 1 identifies thirteen metrics linked to the customer needs. Each metric includes an importance rating and a unit of measurement to help evaluate the performance of the system.

Section 2 develops four analytical models covering motor torque and speed, repeatability error, vibration settling time, and emergency-stop distance and response time. These models use design inputs to calculate predicted performance values, which help determine suitable targets for the system.

Section 3 establishes marginal and ideal targets for all thirteen metrics. These targets are based on the analytical models where applicable, along with competitor comparisons and interview information gathered.

Section 1: Preparing the List of Metrics

The table below presents the engineering metrics that were developed for the prototype based on the customer needs identified. Each metric is linked to a specific customer need and includes an importance rating and unit of measurement to help evaluate the system's performance against the project requirements.

Metric #	Need #	Metric	Imp.	Unit
1	N1	Pass-to-pass repeatability	5	List
2	N2	Maximum carriage speed	4	m/s
3	N2	Maximum carriage acceleration	4	m/s^2
4	N2	Residual vibration settling time after a stop	3	s
5	N3	Pan and tilt range of motion of camera	3	°
6	N4	Ease of programmability	2	List
7	N5	Camera feature trigger latency	2	ms
8	N6	Emergency-stop response time	5	ms
9	N6	Stopping distance to obstacle from maximum speed	5	mm
10	N7	System setup	1	min
11	N8	Prototype bill-of-materials cost	1	$
12	N9	Accuracy of desired movement	4	mm

1.1 Metrics of customer Needs

Metrics 2, 3, and 4 are related to N2, which is regarding the fast and smooth camera movement with minimal vibration. Maximum carriage speed measures how quickly the camera can move, while maximum carriage acceleration measures how quickly it can reach a desired speed. Residual vibration settling time measures how long vibrations take to settle after the carriage stops. Together, these metrics provide a more complete assessment of the system's motion performance for high-speed product cinematography.

Metrics 8 and 9 are related to N5, which concerns safe system operation. Camera feature trigger latency measures the delay between a trigger command and the camera responding, while emergency-stop response time measures how quickly the system responds to an emergency-stop command. Stopping distance from maximum speed provides an additional safety measure by determining how far the carriage travels before coming to a complete stop. These measurements help evaluate the system's ability to coordinate camera functions and stop movement safely.

Metric 13 addresses N9, the accuracy of the desired camera movement. This metric measures the deviation between the programmed movement and the actual movement, expressed in millimetres. It complements the pass-to-pass repeatability metric associated with N1: repeatability evaluates whether the system can reproduce the same movement consistently across multiple attempts, whereas accuracy evaluates how closely the actual movement follows the intended position or path. Both are important for producing controlled camera movements and aligning repeated shots during post-production.

1.2 Needs-to-Metrics Mapping

The figure below shows the relationship between the primary need statements and the metrics, in the same dot-matrix format. The solid dot marks the metric that most directly relates to the need; a hollow dot marks a secondary relationship.

		Metrics											
	Need	Pass-to-pass repeatability	Maximum carriage speed	Maximum carriage acceleration	Residual vibration settling time after a stop	Pan and tilt range of motion of camera	Ease of programmability	Camera feature trigger latency	Emergency-stop response time	Stopping distance to obstacle from maximum	System setup	Prototype bill-of-materials cost	Accuracy of desired movement
1	The camera system repeats programmed camera paths each time.	●			o								
2	The camera system provides smooth camera movement without sudden acceleration or deceleration.	o	●	●	●								
3	The camera system maintains the desired camera orientation during movement.					●							
4	The camera system allows the operator to program and preview camera movements before execution.						●						
5	The camera system provides centralized control of camera movement and connected equipment.							●					
6	The camera system provides immediate and effective safety stopping capabilities during operation.								●	●			
7	The camera system uses affordable, replaceable components to simplify maintenance and repairs.										●		
8	The camera system can be quickly set up and operated in different spaces without requiring a dedicated stage.											●	
9	The camera system can perform programmed movements very accurately with little to no deviation from set path.												●

Section 2: Develop Technical Models of the Product

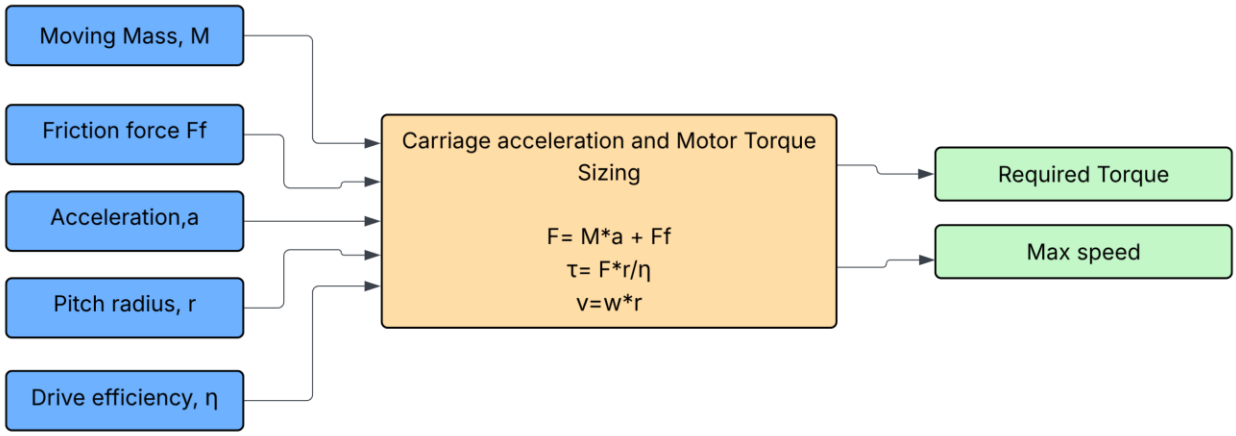
A technical model predicts the value of one or more metrics from a set of design decisions, before any physical prototype is built. Hence, four analytical models are developed below, each addressing metrics that are purely mechanical or control-system quantities and therefore well suited to hand calculation at this stage. The remaining metrics (pan speed, programming time, trigger latency, setup time, and cost) are software, workflow, or procurement quantities that are set directly from the competitive benchmarks and interview evidence.

2.1 Model 1: Carriage Acceleration and Motor Torque Sizing

This model checks whether a realistically sized motor can deliver the target torque and speed for needed to move the camera and take good scene videos/pictures.

Assumed inputs include Moving mass, target acceleration, linear-bearing friction, pitch radius and drive efficiency.

Governing equations includes the force needed to accelerate the carriage against friction, torque required at the motor shaft and carriage speed relating to motor angular speed are the outputs being calculated.

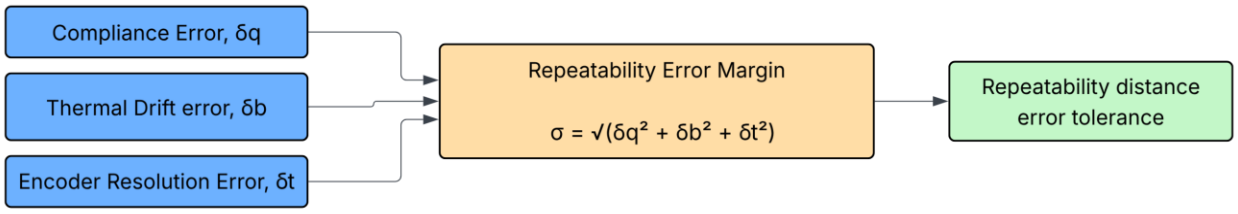


2.2 Model 2: Repeatability Error Budget

This model estimates the best repeatability the system can achieve with a certain margin of error in terms of distance, given plausible component error sources.

Assumed inputs include linear encoder quantization δq, compliance δb and thermal drift over a shooting session δt.

Governing equation includes each error combines by root-sum-square to produce the tolerance range for deviation in repeatability.

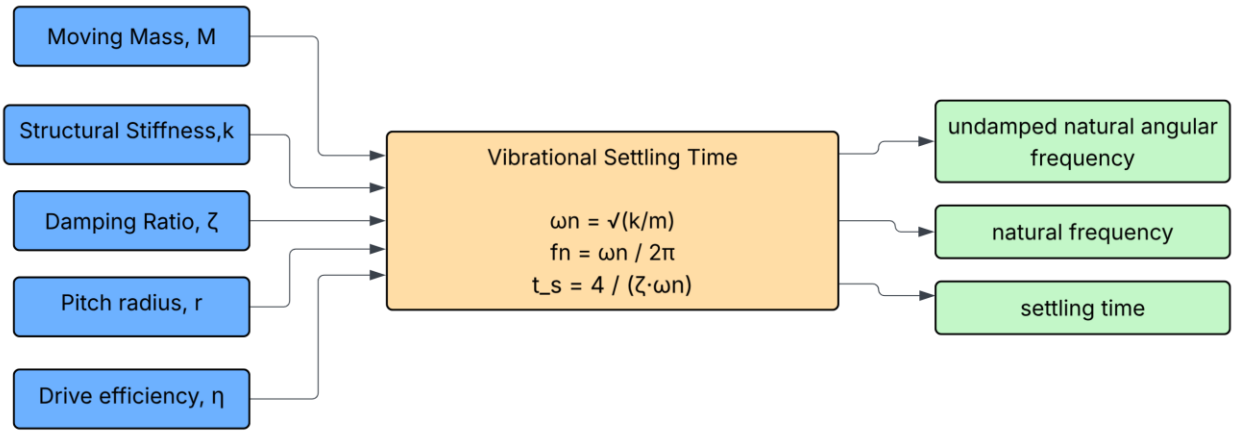


2.3 Model 3: Vibration Settling Time

This model predicts how long the carriage keeps shaking after a fast move stop using a standard second-order mass-spring-damper concept of the rail structure.

Assumed inputs. Moving mass, structural stiffness, damping ratio, spring-damper isolation feet for improved damping design.

Governing equations includes the undamped natural frequency, natural frequency and band settling time are being combined to calculate primarily how long it takes the carriage to stop vibrating after quick movements.

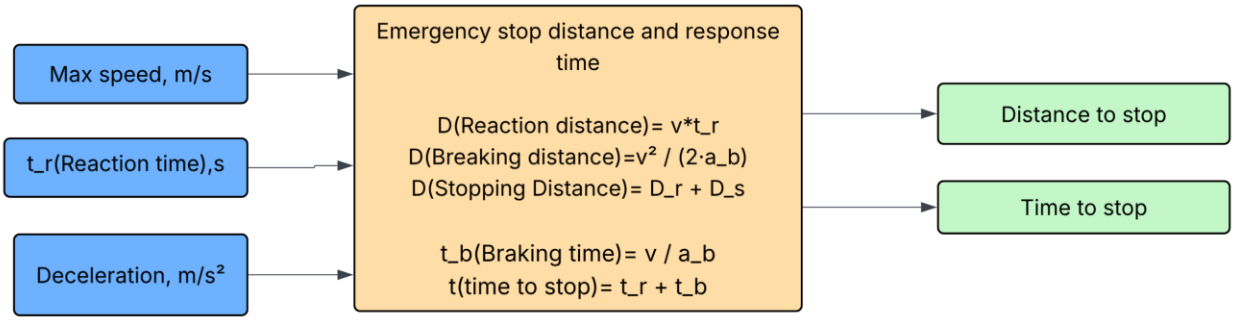


2.4 Model 4: Emergency-Stop Distance and Response Time

This model estimates how far and how quickly the carriage can be brought to a safe stop.

Assumed inputs include maximum carriage speed, sensor-to-drive reaction latency and controlled-stop deceleration.

Governing equations include distance travelled during the reaction delay, distance travelled while decelerating, total stopping and the corresponding times for braking.



Section 3: Establishing Ideal and Marginally Acceptable Target Values

The table below show what the ideal values are in terms of the best result the team could hope for, while the marginally acceptable value is the threshold the product must at least clear to remain viable for its intended use.

Metric #	Need #	Metric	Imp.	Units	Marginal	Ideal
1	N1	Pass-to-pass repeatability	5	List	<0.05 %	<0.01 %
2	N2	Maximum carriage speed	4	m/s	> 1.0	> 2.0
3	N2	Maximum carriage acceleration	4	m/s²	> 2.0	> 4.0
4	N2	Residual vibration settling time after a stop	3	s	< 0.8	< 0.3
5	N3	Pan and tilt range of motion of camera	3	°	> 120	> 180
6	N4	Ease of programmability	2	List	Basic shot programming	Intuitive, code-free programming and preview
7	N5	Camera feature trigger latency	2	ms	< 50	< 10
8	N6	Emergency-stop response time	5	ms	< 300	< 150
9	N6	Stopping distance to obstacle from maximum speed	5	mm	< 300	< 150
10	N7	System setup	1	min	< 20	< 10
11	N8	Prototype bill-of-materials cost	1	$	< 3,000	< 1,500
12	N9	Accuracy of desired movement	4	mm	< 1.0	< 0.5

Reflection and Justification of Marginal and Ideal Metric Values for the CMR

Metric 1: Pass-to-pass repeatability (N1)

Marginal: Less than 0.05% | **Ideal:** Less than 0.01%

The marginal target of less than 0.05% allows a small amount of variation between repeated camera movements. This is important because the system needs to repeat the same path when recording multiple takes so they can line up during editing. The ideal target of less than 0.01% would provide greater consistency and reduce the need for corrections during post-production.

Metric 2: Maximum carriage speed (N2)

Marginal: Greater than 1.0 m/s | **Ideal:** Greater than 2.0 m/s

The marginal target of greater than 1.0 m/s allows the camera to move at a suitable speed for different types of product shots. The ideal target of greater than 2.0 m/s would allow faster movements and provide more flexibility when creating shots. However, the maximum speed will also depend on the camera payload, rail length, motor torque, and vibration produced during movement.

Metric 3: Maximum carriage acceleration (N2)

Marginal: Greater than 2.0 m/s² | **Ideal:** Greater than 4.0 m/s²

The marginal target of greater than 2.0 m/s² allows the carriage to reach its desired speed within a reasonable amount of time. The ideal target of greater than 4.0 m/s² would support faster and more dynamic camera movements. However, higher acceleration could also introduce vibration, belt stretch, and additional stress on the mechanical components, which would need to be considered during the design.

Metric 4: Residual vibration settling time after a stop (N2)

Marginal: Less than 0.8 s | **Ideal:** Less than 0.3 s

The marginal target of less than 0.8 seconds allows the camera system to settle within a reasonable time after a movement ends. This is important because residual vibration can affect image quality, especially during close-up product shots and high-speed cinematography. The ideal target of less than 0.3 seconds would allow the system to settle faster before the next movement or shot begins.

Metric 5: Pan and tilt range of motion of camera (N3)

Marginal: Greater than 120° | **Ideal:** Greater than 180°

The marginal target of greater than 120° provides a useful range of motion for changing the camera angle and following a product. The ideal target of greater than 180° would allow more complex

movements, such as orbiting around a product. A wider range of motion would give the operator more flexibility when creating shots without having to reposition the entire robot.

Metric 6: Ease of programmability (N4)

Marginal: Basic shot programming | **Ideal:** Intuitive, code-free programming and preview

The marginal target allows users to program basic camera movements and repeat them when needed. The ideal target would provide a simple interface where users can set keyframes, adjust movement settings, and preview the camera path without writing code. This would make the CMR easier to use for operators without programming experience and help reduce setup errors before running a movement.

Metric 7: Camera feature trigger latency (N5)

Marginal: Less than 50 ms | **Ideal:** Less than 10 ms

The marginal target of less than 50 milliseconds allows the system to send a trigger signal with a short delay after receiving a command. The ideal target of less than 10 milliseconds would reduce this delay and improve coordination between camera movement and image capture. This is important when the camera needs to trigger a feature or capture an image at a specific point during a programmed movement.

Metric 8: Emergency-stop response time (N6)

Marginal: Less than 300 ms | **Ideal:** Less than 150 ms

The marginal target of less than 300 milliseconds sets a limit on how quickly the system begins responding after the emergency stop is activated. The ideal target of less than 150 milliseconds would reduce this response time. A faster response would help limit unwanted movement and reduce the risk of damage to the camera, robot, or nearby equipment.

Metric 9: Stopping distance to obstacle from maximum speed (N6)

Marginal: Less than 300 mm | **Ideal:** Less than 150 mm

The marginal target of less than 300 mm limits how far the carriage travels before coming to a complete stop when moving at maximum speed. The ideal target of less than 150 mm would allow the system to stop within a shorter distance. This is important for reducing the risk of collisions with nearby objects and ensuring the carriage can stop safely within its operating space.

Metric 10: System setup (N7)

Marginal: Less than 20 minutes | **Ideal:** Less than 10 minutes

The marginal target of less than 20 minutes allows the operator to prepare the system for filming within a reasonable amount of time. This includes mounting the camera, checking connections, and preparing the control system. The ideal target of less than 10 minutes would reduce setup time and

make the CMR more convenient when changing products or camera angles during a filming session.

Metric 11: Prototype bill-of-materials cost (N8)

Marginal: Less than $3,000 | **Ideal:** Less than $1,500

The marginal target of less than $3,000 sets a cost limit for building a functional prototype using suitable motors, linear rails, controllers, and other components. The ideal target of less than $1,500 would make the prototype more affordable while still meeting the main project requirements. Keeping the cost low is important because the project aims to develop a functional camera motion system within a limited budget.

Metric 12: Accuracy of desired movement (N9)

Marginal: Less than 1.0 mm | **Ideal:** Less than 0.5 mm

The marginal target of less than 1.0 mm limits the difference between the programmed camera position and the actual measured position. This is important because the camera needs to follow the intended path accurately when recording product shots. The ideal target of less than 0.5 mm would provide greater precision for close-up filming and repeated movements that need to align across multiple takes during post-production.

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